ST2187 Business Analytics, Applied Modeling and Prediction Coursework

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Executive Summary

This report analyses the Superstore dataset across four years (2020 to 2023), to discover insights to support business decisions related to sales performance, product mix by categories, customer behaviour trends and shipping operations. The goal is to pinpoint key trends, patterns, and opportunities for improvement to drive profitability, enhance operational efficiency and elevate customer satisfaction.

Key Findings:

- Consistent increase in sales over the years. Maintain adequate stock to meet demand (forecasted demand in 6. Forecasts)
- Investigate areas making a loss (e.g. Country: Turkey, City: Lagos)
- For Furniture, re-evaluate pricing and raise profit margin: minimise production and shipping costs by localising suppliers and distributors
- Increase focus on well-performing markets (APAC, EU, US), cities (NYC, LA) & sub-categories (Copiers, Phones)
- Suggestions for Customer Segments:
    - Consumer (highest sales and variability): Personalised marketing for high spenders, promote frequently paired products
    - Corporate: Bulk discounts to incentivise bigger purchases
    - Home Office: Affordable bundles like office supplies necessities
- Streamline shipping operations, merge First & Second Class into “Express Shipping: 1-2days”, reduce shipping delays in Africa
- Review product with excessive discounts to increase profitability

By adopting these recommendations, Superstore can increase overall profitability, streamline operations and improve customer satisfaction. Addressing issues with specific categories and regions/cities, while optimising shipping processes will contribute to a stronger performance.

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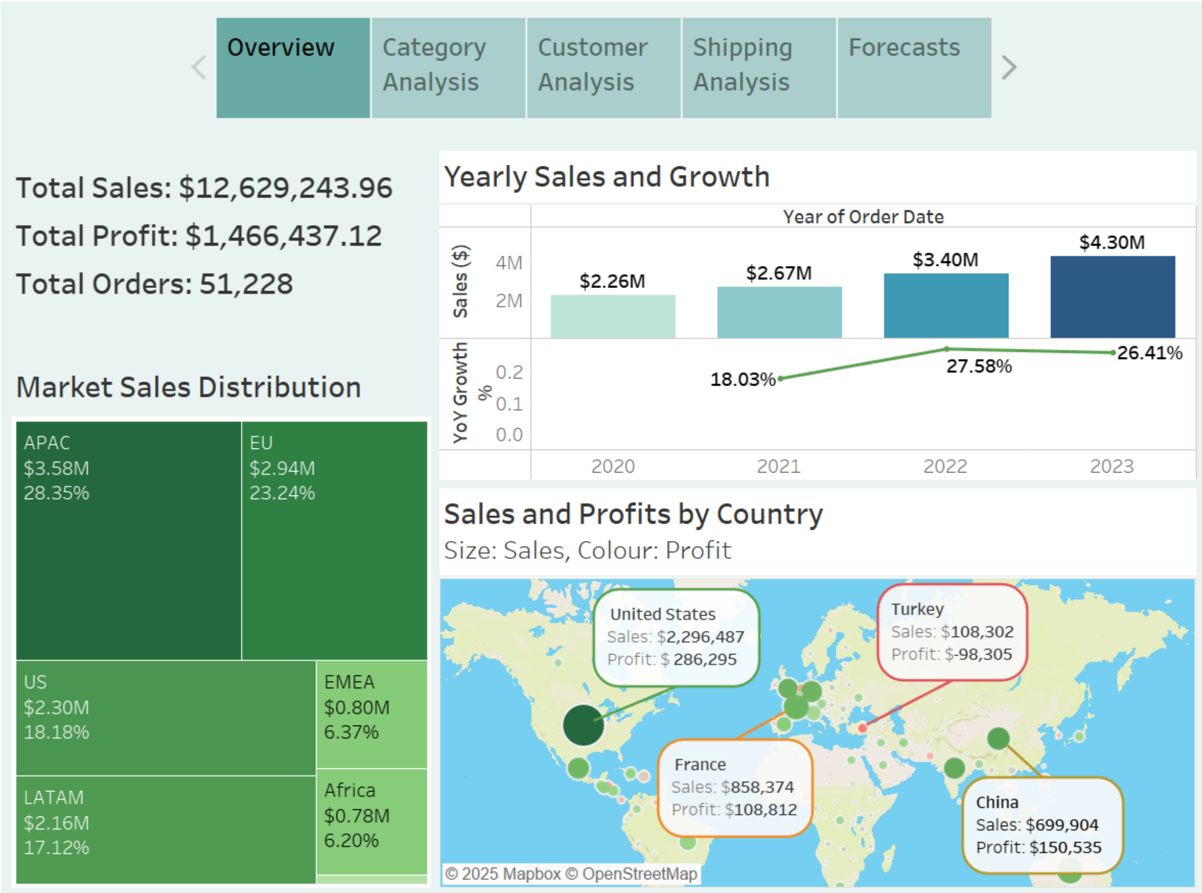
1. Introduction

This business analysis report examines a four-year dataset (2020 – 2023) from Superstore, a global retail company offering a variety of products. In this report, we will analyse Superstore’s sales trends, product categories, customer profiles and shipping metrics. Additionally, it includes a forecast of category sales for the next five quarters and a What-If Analysis on a 40% discount reduction.

Each dashboard’s insights will be elaborated upon in the following sections, accompanied by actionable business recommendations. This will aid senior management in making well-informed, strategic business decisions. With appropriate initiatives, Superstore can close performance gaps and meet its goals by boosting profits, improving operational efficiency and enhancing customer satisfaction.

2. Overview



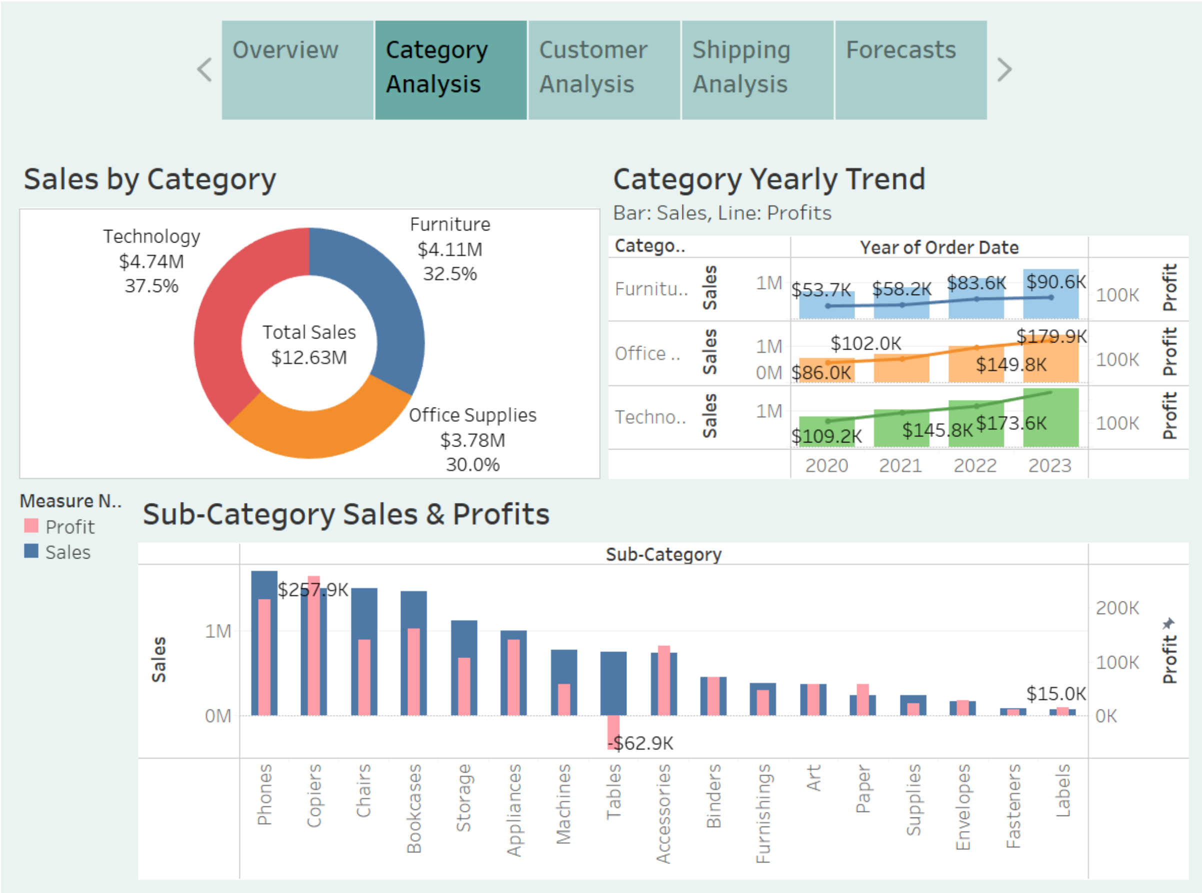
This dashboard shows the business overview: sales, profits and a broad geographical view. There were 51,228 orders across four years, from 2020 to 2023. Superstore's total sales add up to $12,629,243.96, and they earned a total profit of $1,466,437.12.

The company saw a consistent increase in sales over the years. From 2021 to 2021, there was 27.68% increase from the previous year. In 2022, sales growth slowed slightly (26.41%), but it is unlikely a significant cause for concern.

The APAC market contributed most to total sales. The top 2 markets: APAC and EU already account for more than half of total sales (51.59%). There is expansion potential for the smaller markets, either through increasing their market presence or procuring more customers from those areas.

The map shows that US has the highest sales ($2,296,487) and profits ($286,295). France and China follow in terms of sales, with $858,374 and $699,904, respectively. However, although France has higher sales, its profits ($108,812) were lower than China's ($150,5235). The most obvious outlier in the map is Turkey: it suffered a significant loss of $98,305, despite having $108,302 total sales. Perhaps this is due to operational inefficiencies driving up costs, and we recommend Superstore to investigate the causes. Another possibility is profitability challenges due to specific product categories, which we will delve into next.

3. Category Analysis

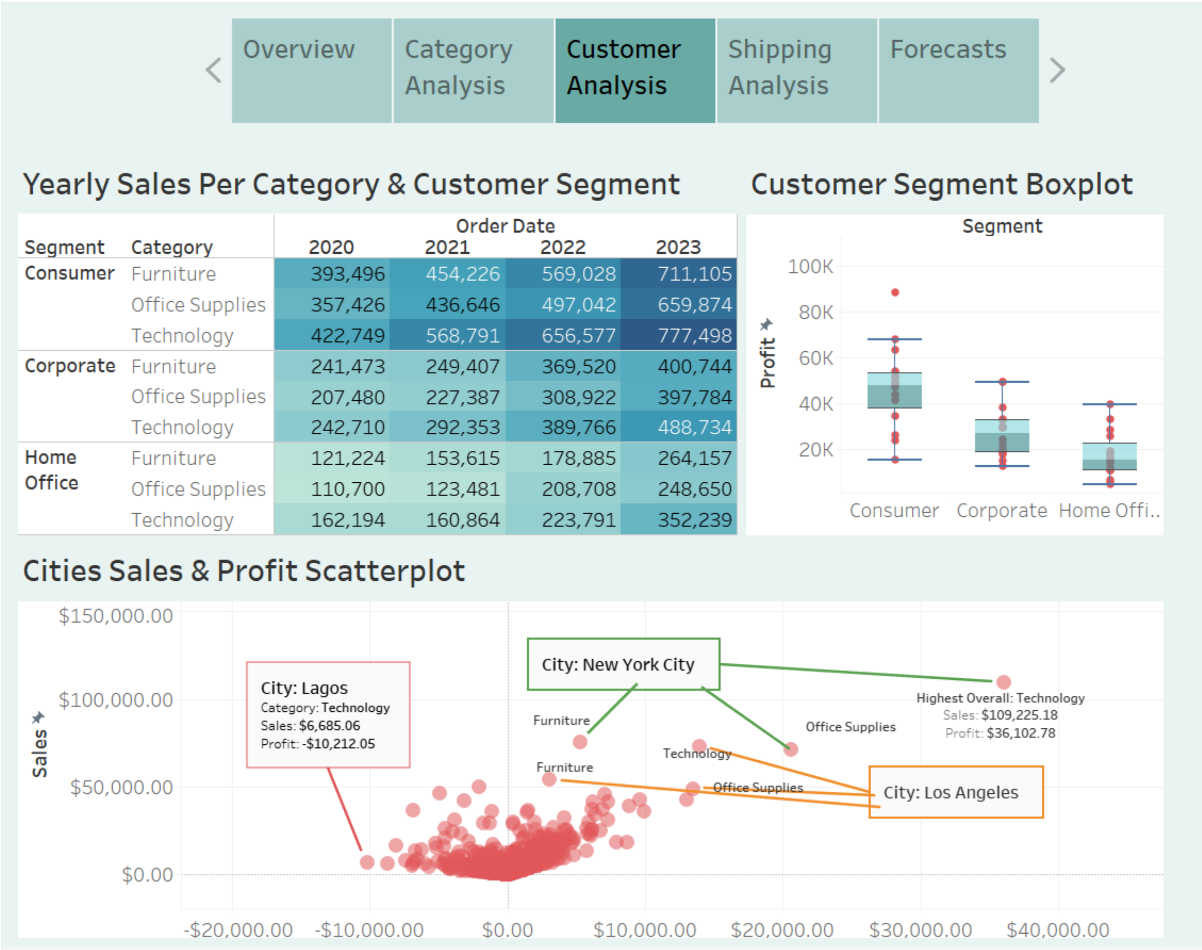


This dashboard is about the product categories: Technology, Office Supplies and Furniture. According to the pie chart, the total sales was distributed somewhat evenly among the three categories. The highest sales are Technology (37.5%), then Furniture (32.5%) and Office Supplies (30.0%).

Examining the yearly trend, Furniture consistently generates the lowest profits, often bringing in almost half the quarterly profits of the other categories. For furniture, the gentler steepness of its line graph also indicates a slow increase in profits, though sales go up. Furniture is likely to be priced too low, affecting its profit potential; Superstore should re-evaluate its prices.

The Sub-Category Sales and Profits graph clearly shows that Tables are making the highest loss of $62.9k. Supply or production chain costs could be high for Tables, affecting profits even if the sales volume is healthy. Given that pricing for Tables generally follow a standard, we recommend not to increase prices significantly. Instead, Superstore should focus on concerns like sourcing suppliers that can provide better prices and identifying any supply chain kinks. Besides that, this graph show that Copiers have the highest profits ($257.9k) followed by Phones. Superstore might be able to increase profits from these sub-categories by expanding their model offerings and keeping up with new releases.

4. Customer Analysis



This dashboard looks at customer segment: Consumer, Corporate, Home Office, and cities' sales and profits. Analysing the Yearly Sales, we see that Consumer brings in highest sales over the years, with Technology-2023 the highest at $777,498.

The boxplot helps us understand the variability of sales volume within each segment. Consumer has the highest median and larger variability, meaning consumers' spending habits differ greatly, notably the much higher red dot outlier. We recommend Superstore to introduce personalised marketing to target the high-spending consumers, and find products frequently purchased together to market them.

From the boxplot, we also see that Corporate and Home Office have more consistent spending. To appeal to them, Superstore can offer corporate customers bulk discounts, incentivising bigger purchases and offer home office customers affordable bundles like office supplies necessities. Corporate and Home Office segments have more consistent spending. To appeal to these segments, Superstore can offer corporate customers bulk discounts to incentivise bigger purchases and offer home office customers affordable bundles such as office supplies necessities.

From the cities scatterplot, we can identify that New York City is the best performing city in terms of sales and profits, followed by Los Angeles, for all product categories. Superstore is advised to invest additional resources in these cities, maintaining sufficient stock for well-performing products. Conversely, Lagos has the worst performance, making a loss of $10,212.05 This could be attributed to Lagos' low profit margin product mix, competition from more localised companies and external factors. Superstore should conduct a thorough analysis of the sales data and implement cost-saving measures.

5. Shipping Analysis

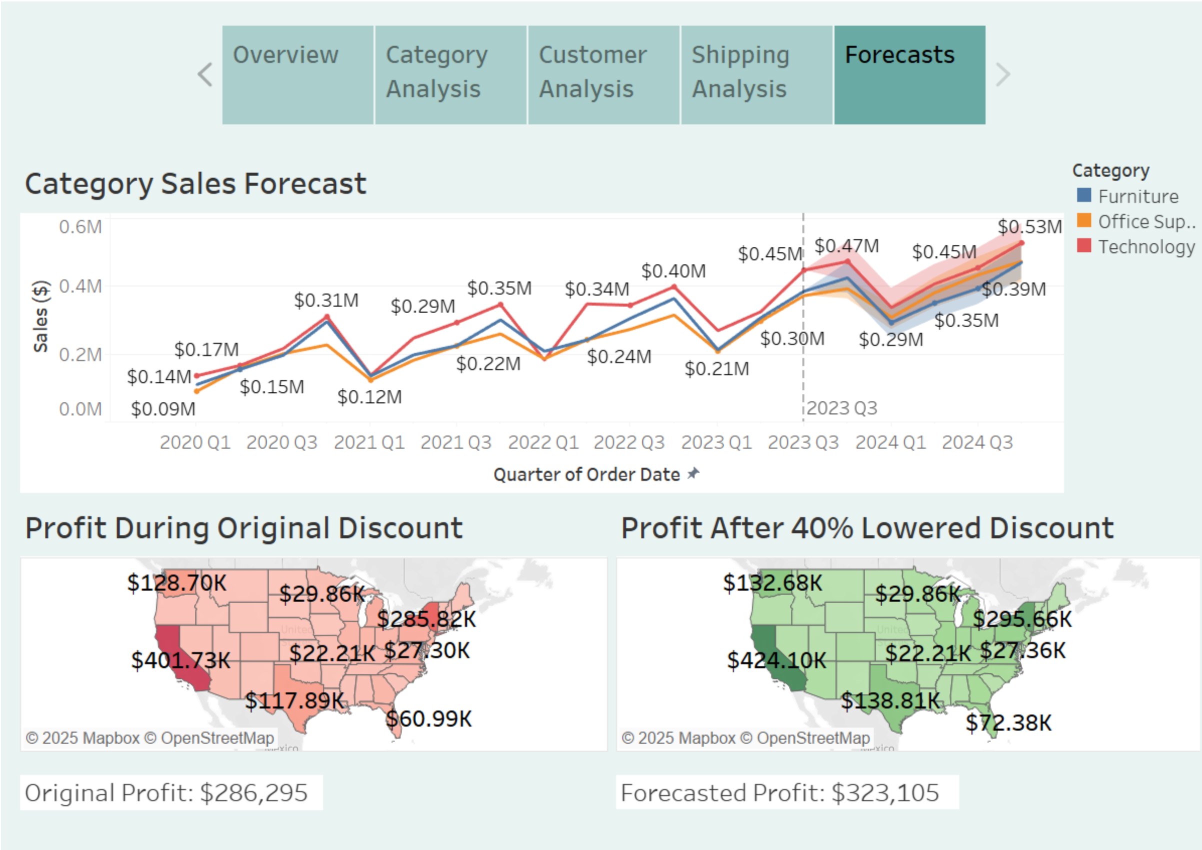
The heatmap visualises the relationship between order priority and shipping mode selected. As order priority increases, customer choose faster shipping modes, especially seen in the absence of orders in Critical Priority – Standard Class. Only 356 out of 1971 Critical priority orders (18%) selected Same Day delivery, likely due to the higher costs. The darkest rectangle appears in Standard Class – Medium at 10,632 orders, as most orders are Medium priority with Standard Class being the most popular option. For Low priority orders, customers only selected Standard Class. Studying this pattern allows us to understand of the impact of urgency on shipping mode preferences.

To increase profitability, the company should combine First Class and Second Class into "Express Shipping: 1-2 days", priced slightly higher than the current average. This simplifies shipping options for customers. For Superstore, it decreases administrative complexity. With consolidated shipping volume for Express Shipping orders, the company may potentially negotiate better rates with shipping providers.

Analysing the scatterplot, we find Tables are the most expensive to ship on average ($92.76), likely explained by its weight and bulkiness. In addition to recommendations regarding Tables previously, we urge Superstore to source more localised suppliers across different regions, to lower shipping costs, which also helps improve the currently negative profit margin of -24.1%. While paper has the highest profit margin at 19.7% at low shipping cost $7.53, it contributes lesser to overall profitability due to its low production and selling price than other products. Instead, Superstore should focus on Copiers and Phones with manageable shipping costs and relatively healthy profit margins.

The bar graph shows that Namibia (7 days), Djibouti and Togo face the longest shipping time for Standard Class. Superstore’s dataset is large and contains outliers, median shipping time is selected instead of average. Customers from these countries may be unsatisfied with the shipping delay, causing loss of customers. Superstore should improve their distribution centres in Africa.

6. Forecasts



Here, we forecasted the sales for the next five quarters. Like the observed trend in Q1 over the past four years, a dip is projected for 2024 Q1. Sales projection for all categories looks optimistic, Superstore can anticipate this sales volume by ensuring adequate stock is available.

A What-If Analysis of US is conducted; we observe that when discounts are lowered by 40%, profits increase from $286,295 to $323,105. We recommend that Superstore reviews products with excessive discounts, to drive higher profits.

7. Conclusion

In this report, we analysed trends across key dimensions including product performance and shipping patterns. Our findings indicate that Superstore should revise their pricing strategy, in terms of improving profit margins for categories like tables, and overall discounts. To ensure customer satisfaction, steps can be taken to improve the shipping process, like addition of “Express Delivery: 1-2Days”.

Although this study offers valuable insights, it is crucial to recognize that external factors also influence Superstore’s performance. We suggest Superstore to conduct market research to better understand their customers and prospects, through questionnaires and focus groups, to improve product offerings. Moving forward, Superstore should leverage these actionable insights to tweak its strategies and achieve sustainable growth.